

Calculation of RR Photon Spectrum including Radiative Cascades in `hydrocal/src/RRphotons.cxx`

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1 Problem

When a free electron is captured by a hydrogenic ion via radiative recombination (RR), two kinds of photons are emitted:

1. **Free-bound (RR) photon:** emitted during the capture of a free electron with kinetic energy E_{kin} and angular momentum l' into a bound state $|n, l\rangle$ with $l = l' \pm 1$.
2. **Cascade photons:** emitted as the excited ion decays through a chain of dipole transitions down to the ground state.

The goal is to compute a complete line list—photon energies, intensities, and natural linewidths—that can later be convolved with a Gaussian instrumental profile using the existing `sum_of_peaks` tool.

2 Free-Bound (RR) Photons

2.1 Photon Energy

The RR photon energy in the ion rest frame is determined by energy conservation. For high- z ions, relativistic (Dirac) energies are used instead of the non-relativistic Bohr formula. The Dirac binding energy of a hydrogenic state $|n, l\rangle$ is computed by averaging over the two fine-structure levels $j = l - 1/2$ and $j = l + 1/2$ with statistical weights $(2j+1)$:

$$E_{\text{Dirac}}(n, l) = \frac{2l E(n, l, j=l-\frac{1}{2}) + 2(l+1) E(n, l, j=l+\frac{1}{2})}{4l + 2} \quad (1)$$

(for $l = 0$ only $j = 1/2$ exists). Each component is given by:

$$E(n, l, j) = m_e c^2 \left(\frac{1}{\sqrt{1 + \kappa^2}} - 1 \right) \quad (2)$$

where

$$\kappa = \frac{z \alpha}{n - k + \sqrt{k^2 - (z \alpha)^2}} \quad (3)$$

with α the fine-structure constant, $k = j + 1/2$, and the result averaged over the two allowed j values for $l > 0$. The RR photon energy is then:

$$E_{\text{RR}}(n, l) = E_{\text{kin}} - E_{\text{Dirac}}(n, l) \quad (4)$$

where $E_{\text{Dirac}}(n, l) < 0$ is the (negative) binding energy. The energy now depends on both n and l due to fine-structure splitting, lifting the l -degeneracy of the non-relativistic formula.

2.2 Line Strength

The RR cross section $\sigma_{\text{RR}}(n, l, E_{\text{kin}})$ is obtained from the quantum-mechanical calculation implemented in `sigmarrqm()` [1]. The line strength of each RR photon equals the cross section for capture into (n, l) :

$$S_{\text{RR}}(n, l) = \sigma_{\text{RR}}(n, l, E_{\text{kin}}) \quad [\text{cm}^2] \quad (5)$$

The output lists one entry per (n, l) state (not grouped by n), so the l -distribution of the initial capture is preserved.

2.3 Natural Linewidth

The RR photon has no natural broadening from the final state lifetime (the photon energy is fixed by energy conservation), so the natural width is set to zero. Any broadening comes from the experimental electron energy spread and is applied later via convolution.

2.4 Label Convention

Each RR line is labelled $(E, l-1)/(E, l+1) \rightarrow (n, l)$ for $l > 0$, or $(E, 1) \rightarrow (n, 0)$ for $l = 0$, indicating the allowed free-electron angular momenta consistent with the dipole selection rule $\Delta l = \pm 1$. The letter “ E ” represents the continuum electron. For a relativistic target state ($n \leq n_{\text{rel}}$) the label additionally carries $2j$: $(E, l-1)/(E, l+1) \rightarrow (n, l, 2j)$; the continuum partial waves then correspond to the E1-coupled channels $\kappa_c = \kappa \pm 1$ and $\kappa_c = -\kappa$.

3 Cascade Photons

3.1 Rate Matrix

The radiative cascade is governed by the coupled rate equations

$$\frac{dp_i}{dt} = \sum_j M_{ij} p_j \quad (6)$$

where p_i is the population of state i (index $i = n(n-1)/2 + l$) and $\mathbf{M}$ is the rate matrix:

$$M_{ii} = -\frac{1}{\tau_i}, \quad M_{ji} = \frac{B(i \rightarrow j)}{\tau_i} \quad (j \neq i) \quad (7)$$

with τ_i the radiative lifetime of state i and $B(i \rightarrow j)$ the branching ratio for the transition $i \rightarrow j$ (the sum of the E1, E2, and M1 rates; parity keeps E1 separate from E2/M1). The ground state $(1, 0)$ has $\tau = \infty$ (no decay), so its column in $\mathbf{M}$ is zero. Among relativistic states ($n \leq n_{\text{rel}}$) each fine-structure level j is a separate state, and E1 as well as E2/M1 transitions enter the rate matrix as individual entries $M_{ji} = A(i \rightarrow j)$.

3.2 Time-Integrated Populations

The total photon yield from each transition requires the time-integrated population of the upper state:

$$\bar{p}_i = \int_0^\infty p_i(t) dt \quad (8)$$

Since the ground state makes $\mathbf{M}$ singular, we restrict to the excited-state submatrix $\mathbf{M}_{\text{exc}}$ of dimension $N_{\text{exc}} = n_{\text{max}}(n_{\text{max}}+1)/2 - 1$ (all states except $(1, 0)$). Then:

$$\bar{\mathbf{p}}_{\text{exc}} = -\mathbf{M}_{\text{exc}}^{-1} \mathbf{p}_0 \quad (9)$$

where $\mathbf{p}_0$ is the initial population vector (RR cross sections) and $\mathbf{M}_{\text{exc}}$ is invertible because all its eigenvalues have negative real parts.

The linear system

$$\mathbf{M}_{\text{exc}} \bar{\mathbf{p}}_{\text{exc}} = -\mathbf{p}_0 \quad (10)$$

is solved by Gaussian elimination with partial pivoting (`matrixSolve()`).

3.3 Photon Energy

For each dipole-allowed transition $(n_1, l_1) \rightarrow (n_2, l_2)$ with $n_2 < n_1$ and $l_2 = l_1 \pm 1$:

$$E_\gamma = E_{\text{Dirac}}(n_1, l_1) - E_{\text{Dirac}}(n_2, l_2) \quad (11)$$

where $E_{\text{Dirac}}(n, l)$ is given by Eq. (1). For high- z ions the fine-structure splitting of both states is taken into account, so cascade photon energies differ from the non-relativistic Balmer/Ritz formula.

3.4 Line Strength

The total number of photons emitted in transition $i \rightarrow j$, integrated over all time, is:

$$S(i \rightarrow j) = A(i \rightarrow j) \bar{p}_i \quad (12)$$

where $A(i \rightarrow j) = \text{trans}(n_1, l_1, n_2, l_2)$ is the Einstein A coefficient (transition rate in s^{-1}) and $\bar{p}_i$ is the time-integrated population of the upper state in units of $\text{s}\cdot\text{cm}^2$. The product has units of cm^2 .

3.5 Natural Linewidth

The natural Lorentzian FWHM of a spectral line is determined by the lifetimes of both the upper and lower states:

$$\Gamma_{ij} = \hbar \left(\frac{1}{\tau_i} + \frac{1}{\tau_j} \right) \quad (13)$$

For transitions to the ground state ($n_2 = 1$), $\tau_j \rightarrow \infty$ and the width reduces to $\Gamma = \hbar/\tau_i$.

3.6 Label Convention

Each cascade line is labelled $(n_1, l_1) \rightarrow (n_2, l_2)$ with the actual quantum numbers. For relativistic states ($n \leq n_{\text{rel}}$) the label is j -resolved, $(n_1, l_1, 2j_1) \rightarrow (n_2, l_2, 2j_2)$, and the rate is $A_{\text{total}} = A_{\text{E1}}$ or $A_{\text{E2}} + A_{\text{M1}}$ depending on the selection rules. E2/M1 lines (e.g. the fine-structure transition $2p_{3/2} \rightarrow 2p_{1/2}$) therefore appear as individual spectral lines at their characteristic energy.

4 Relativistic treatment for $n \leq n_{\text{rel}}$

For high- z ions (or when $n_{\text{rel}} > 0$) the states with $n \leq n_{\text{rel}}$ are described by Dirac hydrogenic spinors with a definite total angular momentum j (encoded in κ), while higher- n states keep the non-relativistic treatment of the previous sections. For such states:

- The binding energy is the Dirac value $E(n, \kappa)$; the two fine-structure components of an (n, l) shell are treated as individual states and labelled $(n, l, 2j)$ (e.g. $(2, 1, 1)$ for $2p_{1/2}$ and $(2, 1, 3)$ for $2p_{3/2}$).
- The RR cross section for capture into (n, κ) is obtained from the relativistic Milne relation, summed over the E1-coupled continuum channels $\kappa_c = \kappa \pm 1$ and $\kappa_c = -\kappa$. The direct-RR label is $(E, l-1)/(E, l+1) \rightarrow (n, l, 2j)$.

- Cascade transitions between two relativistic states are j -resolved: the rate is $A_{\text{total}}(n_i, \kappa_i \rightarrow n_j, \kappa_j)$ (sum of E1, E2, and M1; E1 never coincides with E2/M1 because of parity, so the sum equals the allowed multipole), the photon energy is the difference of the two Dirac binding energies, and the E2/M1 transitions enter both the rate matrix and the line list as individual entries.
- Decays from a non-relativistic upper state into a relativistic fine-structure doublet are distributed by statistical weight $(2j+1)/[2(2l+1)]$.

The rate matrix, population solution, and line construction are otherwise unchanged; every transition in the final line list connects states with definite j wherever the lower state is relativistic.

5 Output Format

The output file (`*.RRlines`) is a columnar text file compatible with the existing `read_peak()` / `sum_of_peaks()` infrastructure. Each data line is preceded by a `###` comment line carrying the transition label:

```
#####
### Photon spectrum resulting from the radiative cascade following
### radiative recombination (RR lines + cascade lines)
###
### hydrocal SVN revision : 2089:2090M
### filename : Pb82RRphotons.RRlines
### number of spec. lines : 2680
### start date & time : Fri Jul 31 11:28:21 2026
### nuclear charge Zeff : 82
### electron energy (eV) : 1e-05
### nmax : 20
### min photon energy (eV) : 0
### relativistic for n<= : 0
#####
#### energy (eV) [1] cross_section (cm^2) [2] line-width (eV) [3]
###-----
### (20,0)->(19,1)
2.34962616e+01 6.17478755e-18 1.60586110e-01
### (20,1)->(19,2)
2.36837825e+01 1.11557824e-18 1.78156903e-01
...
```

Lines starting with `#` are skipped by `read_peak()` via the `skip_comment_lines()` function. The three numeric columns are:

1. **Energy** (eV): photon energy in the ion rest frame.
2. **Strength** (cm²): integrated line intensity (cross section for RR lines; transition rate $\times$ time-integrated population for cascade lines).
3. **Width** (eV): natural Lorentzian FWHM (zero for RR lines).

Lines are sorted by photon energy.

6 Convolution

The line list can be convolved with a Gaussian instrumental profile using `sum_of_peaks` (mode 2 = Lorentz). The tool reads the `*.RRlines` file, prompts for a Gaussian FWHM, and produces a broadened spectrum as a function of photon energy. Voigt profiles (Lorentzian $\otimes$ Gaussian) are obtained automatically when the input contains non-zero Lorentzian widths.

7 Implementation Notes

- All cascade computations (fully non-relativistic and mixed relativistic/non-relativistic) are handled by a single code path (`cascade_spectrum_rel()` in `RRphotons.cxx`). For $n_{\text{rel}} = 0$ all states are treated non-relativistically; for $n_{\text{rel}} > 0$ the states with $n \leq n_{\text{rel}}$ are treated with the Dirac rates and the higher- n states with the non-relativistic rates.
- The rate matrix $\mathbf{M}$ carries the individual relativistic E1, E2 and M1 transition rates (from `dirac_total_transrate()`) and the non-relativistic dipole branching ratios (from `RADRATE::branch()`) as its off-diagonal entries. Decays from non-relativistic upper states require $\Delta l = \pm 1$ (dipole selection rule); decays between relativistic states are restricted by the Dirac parity and angular-momentum selection rules.
- The `RADRATE` class provides transition rates, branching ratios, and lifetimes computed from hydrogenic oscillator strengths. The rates are in s^{-1} and the lifetimes in s.
- The linear system $\mathbf{M}_{\text{exc}} \bar{\mathbf{p}} = -\mathbf{p}_0$ is solved by Gaussian elimination with partial pivoting (`matrixSolve()` in `hydromath.cxx`).
- For zero electron energy ($E_{\text{kin}} < 10^{-10}$ eV), the RR cross section is evaluated at $E = 10^{-12}$ eV to avoid the $1/E$ singularity of the Wigner threshold law.
- The `calc_spectrum()` function is accessible from the hydrocal menu as RR cross sections group, option 4.

References

- [1] S. Schippers, Experimente zur Photorekombination Atomarer Ionen an Schwerionenspeicherringen (Habilitation Thesis), Justus-Liebig-Universität Gießen, Germany, 2002, Chapter 3.